ACADEMIC CENTRAL

PRACTICE EXAMINATION

N6

MATHEMATICS N6

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central N6 Mathematics N6 paper 1 · practice a.

Written by Academic Central for practice, following the NATED (Report 191) syllabus for N6. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Give method marks for correct working even if the final answer is wrong (consistent accuracy). Accept equivalent forms of the answers and other correct methods. Penalise a missing constant of integration only once.

QUESTION 1

1.1.1 $2xy^3$ ✓ $- 3\sin y$ ✓ (2)

1.1.2 $3x^2y^2$ ✓ $- 3x\cos y$ ✓ (2)

1.1.3 $6xy^2$ ✓ $- 3\cos y$ ✓ (2)

1.2 $\frac{\partial V}{\partial r} = 2\pi rh = 120\pi$ ✓; $\frac{\partial V}{\partial h} = \pi r^2 = 25\pi$ ✓; $\delta r = 0{,}1$ and $\delta h = -0{,}2$ ✓; $\delta V \approx 120\pi(0{,}1) + 25\pi(-0{,}2)$ ✓ $= 12\pi - 5\pi = 7\pi$ ✓ $\approx 21{,}991\ \text{cm}^3$ (increase) ✓ (6)

[12]

QUESTION 2

2.1 x^3 ✓ $- 4\ln|x|$ ✓ $+ \frac{1}{2}e^{2x} + C$ ✓ (3)

2.2 $\frac{1}{2}\int(\sin 5x + \sin x)\,dx$ ✓✓ $= -\frac{\cos 5x}{10}$ ✓ $- \frac{\cos x}{2} + C$ ✓ (4)

2.3 $u = x,\ dv = e^{3x}dx,\ v = \frac{1}{3}e^{3x}$ ✓; $= \frac{x}{3}e^{3x} - \int\frac{1}{3}e^{3x}\,dx$ ✓ $= \frac{x}{3}e^{3x}$ ✓ $- \frac{1}{9}e^{3x} + C$ ✓ (4)

2.4 $\frac{5x+1}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2}$ ✓; $5x + 1 = A(x+2) + B(x-1)$ ✓; $x = 1$: $A = 2$ ✓; $x = -2$: $B = 3$ ✓; $\int = 2\ln|x-1|$ ✓ $+ 3\ln|x+2| + C$ ✓ (6)

2.5 $\int(1 - \sin^2 x)\cos x\,dx$ ✓; let $u = \sin x,\ du = \cos x\,dx$ ✓; $= \sin x$ ✓ $- \frac{\sin^3 x}{3} + C$ ✓ (4)

2.6 $a = 3$ ✓; $\frac{1}{3}\tan^{-1}\frac{x}{3}$ ✓ $+ C$ ✓ (3)

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QUESTION 3

3.1 Sketch showing the parabola, the line and the enclosed area ✓; $x^2 = 2x$ gives $x = 0$ and $x = 2$ ✓; $A = \int_0^2(2x - x^2)\,dx$ ✓ $= [x^2 - \frac{x^3}{3}]_0^2$ ✓ $= 4 - \frac{8}{3} = \frac{4}{3}$ $\approx 1{,}333$ square units ✓ (5)

3.2 $V = \pi \int_0^2 (x^2)^2\ dx\checkmark = \pi \int_0^2 x^4\ dx\checkmark = \pi[\ \frac{x^5}{5}\ ]_0^2\ \checkmark = \frac{32\pi}{5}\ \checkmark \approx 20{,}106$ cubic units ✓ (5)

3.3 $A = \int_0^2 x^2\ dx = \frac{8}{3}\ \checkmark$; $\int xy\ dx = \int_0^2 x^3\ dx = 4\ \checkmark$; $\bar{x} = 4 \div \frac{8}{3}\ \checkmark = 1{,}5\ \checkmark$; ½ $\int y^2\ dx = ½ \int_0^2 x^4\ dx = \frac{16}{5}\ \checkmark\checkmark$; $\bar{y} = \frac{16}{5} \div \frac{8}{3}\ \checkmark = 1{,}2\ \checkmark$; centroid (1,5 ; 1,2) (8)

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QUESTION 4

4.1 Sketch with a horizontal strip of width b and thickness δy at height y ✓; $\delta A = b\,\delta y$ ✓; $\delta I = y^2\ b\,\delta y$ ✓; $I = \int_0^h by^2\ dy\checkmark = b[\ \frac{y^3}{3}\ ]_0^h\ \checkmark = \frac{bh^3}{3}\ \checkmark$ (6)

4.2 $I = \frac{6 \times 4^3}{3}\ \checkmark\checkmark = 128\ \text{cm}^4\ \checkmark$ (3)

4.3 $A = 24\ \text{cm}^2\ \checkmark$; $k = \sqrt{(\frac{128}{24})}\ \checkmark = 2{,}309\ \text{cm}\ \checkmark$ (3)

[12]

QUESTION 5

5.1 $\frac{dy}{y} = 2x\,dx\checkmark$; $\ln y = x^2 + C\ \checkmark\checkmark$; $y = Ae^{x^2}\ \checkmark$; $y(0) = 3$ gives $A = 3\ \checkmark$; $y = 3e^{x^2}\ \checkmark$ (6)

5.2 $P = 2$, $Q = e^{-x}\ \checkmark$; $IF = e^{\int 2\,dx} = e^{2x}\ \checkmark\checkmark$; $ye^{2x} = \int e^{-x} \cdot e^{2x}\ dx\checkmark = \int e^x\ dx\checkmark$ $= e^x + C\ \checkmark$; $y = e^{-x} + Ce^{-2x}\ \checkmark\checkmark$ (8)

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QUESTION 6

6.1 $m^2 - 5m + 6 = 0\ \checkmark$; $(m-2)(m-3) = 0$, $m = 2$ or $3\ \checkmark$; $y = Ae^{2x} + Be^{3x}$ ✓✓ (4)

6.2 $m^2 + 4m + 4 = 0$, $m = -2$ (equal roots) ✓; $y = (A + Bx)e^{-2x}\ \checkmark\checkmark$ (3)

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6.3 Auxiliary equation $m^2 + 2m + 5 = 0$ ✓; $m = \frac{-2 \pm \sqrt{(4 - 20)}}{2} = -1 \pm j2$ ✓; CF: $y = e^{-x}(A \cos 2x + B \sin 2x)$ ✓✓; PI: try $y = C$, then $5C = 10$, $C = 2$ ✓✓; general solution $y = e^{-x}(A \cos 2x + B \sin 2x) + 2$ ✓; $y(0) = 3$: $A + 2 = 3$, $A = 1$ ✓; $\frac{dy}{dx} = -e^{-x}(A \cos 2x + B \sin 2x) + e^{-x}(-2A \sin 2x + 2B \cos 2x)$ ✓✓; at $x = 0$: $-A + 2B = 0$ ✓, $B = ½$ ✓; $y = e^{-x}(\cos 2x + ½ \sin 2x) + 2$ ✓ (13)

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TOTAL: 100

END OF MARKING GUIDELINE